



Drug-Induced Liver Injury (DILI) — MCQ Practice

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What is the ONE feature that most clearly distinguishes intrinsic from idiosyncratic hepatotoxicity?

EASY

- A. Intrinsic is dose-dependent and predictable; idiosyncratic has no dose relationship and is unpredictable
- B. Intrinsic only occurs in children; idiosyncratic only in adults
- C. Intrinsic always requires a liver transplant; idiosyncratic never does
- D. Intrinsic is always fatal; idiosyncratic is always reversible
- E. Intrinsic only affects the biliary tract; idiosyncratic only affects hepatocytes

2. Which drug is the classic example of zonal (centrilobular) necrosis hepatotoxicity, and what is its antidote?

EASY

- A. Isoniazid; rifampin
- B. Acetaminophen; N-acetylcysteine
- C. Amiodarone; ursodeoxycholic acid
- D. Valproic acid; naloxone
- E. Tetracycline; vitamin K

3. What is the single most important first step in managing any case of DILI?

EASY

- A. Start N-acetylcysteine regardless of the causative drug
- B. Remove/discontinue the offending agent
- C. Begin corticosteroid therapy immediately
- D. Perform liver biopsy before any other action
- E. Start ursodeoxycholic acid

4. Which laboratory pattern is characteristic of hepatocellular DILI versus cholestatic DILI?

EASY

- A. Hepatocellular → raised ALP; Cholestatic → raised ALT/AST
- B. Hepatocellular → raised ALT/AST; Cholestatic → raised ALP
- C. Both show identical lab patterns
- D. Neither pattern involves liver enzymes
- E. Hepatocellular → raised bilirubin only; Cholestatic → raised albumin only

5. A 6-year-old with influenza is given aspirin for fever and subsequently develops confusion, vomiting, and elevated ammonia. What is the diagnosis and mechanism?

MEDIUM

- A. Acetaminophen toxicity via NAPQI accumulation
- B. Reye's syndrome — mitochondrial dysfunction causing microvesicular steatosis, triggered by aspirin during viral illness in a child
- C. Isoniazid hepatotoxicity via toxic CYP450 metabolite
- D. Amiodarone toxicity via phospholipidosis
- E. Valproic acid toxicity via impaired β -oxidation

6. A patient recently finished amoxicillin-clavulanate and develops jaundice, pruritus, and markedly raised ALP with only mildly elevated ALT. What is the injury pattern and mechanism?

MEDIUM

- A. Zonal necrosis from direct hepatotoxicity
- B. Cholestatic hepatitis, via clavulanic acid triggering an idiosyncratic hypersensitivity reaction against hepatocytes/bile duct cells
- C. Granulomatous hepatitis from talc exposure
- D. Microvesicular steatosis from mitochondrial fatty-acid oxidation impairment
- E. Acute hepatocellular injury from a toxic CYP450 metabolite

7. A patient on long-term amiodarone develops chronic hepatocellular injury with fibrosis, which persists even after the drug is stopped. What does this specific persistence suggest about amiodarone's mechanism compared to most other DILI drugs?

MEDIUM

- A. Amiodarone accumulates in the liver causing phospholipidosis and mitochondrial dysfunction, and unlike most DILI (which is favourable after discontinuation), amiodarone-induced injury may persist due to this tissue accumulation
- B. Amiodarone injury always fully reverses immediately upon discontinuation, identical to all other DILI drugs
- C. This persistence indicates the diagnosis must be wrong and is unrelated to amiodarone
- D. Amiodarone has no described mechanism of hepatotoxicity
- E. This pattern is identical to acetaminophen's zonal necrosis mechanism

8. A patient develops idiosyncratic hepatotoxicity from a sulfonamide, with onset 2 weeks after starting the drug, accompanied by rash, fever, and joint pain, recurring on rechallenge. Which subtype of idiosyncratic reaction does this represent?

MEDIUM

- A. Metabolic idiosyncratic reaction
- B. Hypersensitivity idiosyncratic reaction — immunologic, faster onset (1-5 weeks), recurs with rechallenge, with rash/fever/joint pain
- C. Intrinsic dose-dependent reaction
- D. Zonal necrosis pattern
- E. Granulomatous reaction from talc

9. A patient started on isoniazid for latent TB develops asymptomatic transaminase elevation on routine monitoring. Considering isoniazid's described mechanism, how should this be interpreted?

MEDIUM

- A. This must be intrinsic, dose-dependent toxicity requiring immediate high-dose N-acetylcysteine
- B. Isoniazid causes idiosyncratic hepatocellular injury via a toxic CYP450 metabolite (acetylhydrazine) with genetic susceptibility, consistent with the observed transaminase elevation, and should be monitored per idiosyncratic-reaction principles rather than assumed to be dose-related
- C. This finding is unrelated to isoniazid and should be ignored entirely
- D. This pattern is diagnostic of Reye's syndrome
- E. Isoniazid causes only cholestatic injury, so ALT elevation would be unexpected

10. A patient on long-term methotrexate for an autoimmune condition develops evidence of hepatic fibrosis on follow-up imaging. What is the described mechanistic pathway linking methotrexate to this finding?

MEDIUM

- A. Methotrexate directly causes zonal necrosis like acetaminophen
- B. Methotrexate raises homocysteine, causing oxidative stress that activates hepatic stellate cells, leading to fibrosis
- C. Methotrexate causes cholestatic injury via bile transporter impairment
- D. Methotrexate causes microvesicular steatosis via impaired fatty-acid oxidation
- E. Methotrexate has no described hepatotoxic mechanism

11. A patient taking sustained-release niacin for dyslipidemia develops elevated aminotransferases with a mixed cholestatic pattern. How does this compare to statin-associated liver findings described in the lecture?

MEDIUM

- A. Statins have a described higher hepatotoxic potential than niacin
- B. Sustained-release niacin has the highest hepatotoxic potential of the antihyperlipidemics, while statins rarely cause significant injury despite sometimes causing asymptomatic transaminase rises
- C. Niacin and statins have identical, negligible hepatotoxic risk
- D. Statins are described as more hepatotoxic than niacin in every scenario
- E. Niacin has no described hepatotoxic risk at all

12. A patient on azathioprine for years develops non-cirrhotic portal hypertension without frank cirrhosis on biopsy. What specific histologic finding is described as underlying this presentation?

MEDIUM

- A. Zonal centrilobular necrosis
- B. Nodular regenerative hyperplasia (NRH)
- C. Granulomatous hepatitis
- D. Microvesicular steatosis
- E. Cholestatic hepatitis

13. A researcher notes that African Americans have a described increased risk of hepatotoxicity mostly from trimethoprim-sulfamethoxazole, while White Americans have increased risk mostly from amoxicillin-clavulanate. What does this pattern illustrate about DILI risk factors?

MEDIUM

- A. Race/ethnicity has no described bearing on DILI risk according to this lecture
- B. Race/ethnicity is one of several described DILI risk factors, with different populations showing different predominant culprit drugs, alongside other factors like age, sex, and genetic CYP polymorphisms
- C. Only genetic factors matter; race/ethnicity is a social construct with zero biological correlation described here
- D. This pattern applies only to intrinsic, not idiosyncratic, hepatotoxicity
- E. This pattern means these two drugs should never be prescribed to any patient regardless of race

14. A patient with tetracycline-associated microvesicular steatosis is compared to a patient with valproic acid-associated steatosis/chronic hepatocellular injury. What shared underlying mechanistic theme connects these two different drugs' hepatotoxicity?

MEDIUM

- A. Both work through an identical CYP450-mediated hypersensitivity mechanism
- B. Both involve impairment of mitochondrial fatty-acid oxidation/ β -oxidation, leading to triglyceride/lipid accumulation in hepatocytes
- C. Both work through NAPQI accumulation, identical to acetaminophen
- D. Both work through bile transporter protein impairment, identical to anabolic steroids
- E. Neither drug has any described hepatotoxic mechanism

15. A patient develops DILI from a drug not explicitly named in this lecture's core table. Using the lecture's described 3-step mechanistic cascade (direct cell stress/ROS during CYP450 oxidation → impaired mitochondrial function → triggered immune reactions), which piece of clinical evidence would MOST strongly suggest the reaction has progressed to the final ("triggered immune reactions") stage rather than remaining at an early metabolic stage?

HARD

- A. Isolated, mild, asymptomatic transaminase elevation alone
- B. Rash, fever, and eosinophilia alongside liver injury, suggesting an immune-mediated component consistent with the hypersensitivity idiosyncratic pattern described elsewhere in the lecture
- C. A simple dose-dependent relationship between drug amount and severity
- D. Elevated ALP alone with no other findings
- E. A brief, predictable latent period with no immune features

16. A clinician is evaluating a patient with elevated transaminases who takes both isoniazid (idiosyncratic, hepatocellular) and sustained-release niacin (intrinsic tendency, mixed pattern) concurrently. Disentangling the causative agent is difficult. Which reasoning approach, grounded in the lecture's described discriminators, would be most methodologically sound?

HARD

- A. Assume it must be isoniazid, since idiosyncratic reactions are always the more likely cause in any concurrent-drug scenario
- B. Systematically apply the described discriminators (dose relationship, predictability, latent period, and characteristic injury pattern/histology) to each drug's known profile, and consider a temporal relationship (e.g., timing relative to when each drug was started or dose-adjusted) rather than assuming causation from one drug alone
- C. Assume it must be niacin, since only intrinsic hepatotoxins can be reliably diagnosed
- D. This scenario is unsolvable and no reasoning approach exists
- E. Simply stop both drugs simultaneously with no further diagnostic reasoning needed

17. A patient who took a large acetaminophen overdose 2 hours ago is being evaluated for N-acetylcysteine therapy. Based on the lecture's described mechanism (glutathione depletion → NAPQI accumulation → zonal necrosis), why is early administration of NAC more effective than late administration, even though NAC "replenishes glutathione and directly detoxifies NAPQI"?

HARD

- A. NAC timing has no bearing on efficacy per the described mechanism
- B. Early administration intervenes before glutathione stores are fully depleted and before NAPQI has accumulated to levels causing irreversible zonal necrosis, whereas late administration means NAC must contend with a larger NAPQI burden and potentially already-established hepatocyte injury
- C. NAC only works if given more than 24 hours after ingestion
- D. NAC has no direct detoxifying action on NAPQI regardless of timing
- E. Early administration is worse because it prevents natural glutathione regeneration

18. A pharmacovigilance analyst is reviewing DILI as "the #1 cited reason for drug withdrawal from the market (up to 32% of withdrawals)." Considering that most DILI cases are described as asymptomatic, what does this combination of facts suggest about how DILI is typically detected in the drug development and post-marketing surveillance process?

HARD

- A. Since DILI is asymptomatic, it must be undetectable and therefore cannot be the leading cause of withdrawals — these two facts are contradictory
- B. Since most DILI is asymptomatic, detection likely depends heavily on routine laboratory monitoring (e.g., liver enzymes) during trials and post-marketing surveillance rather than symptom-triggered reporting alone, which has implications for how thoroughly a drug must be monitored before its hepatotoxic potential becomes apparent
- C. DILI is always symptomatic and jaundice is the way all cases are detected, making this analysis unnecessary
- D. Drug withdrawals are unrelated to any laboratory monitoring and rely solely on patient-reported symptoms
- E. This combination of facts has no practical implication for surveillance practice

19. A patient has anabolic steroid-induced cholestasis that is noted to be "usually reversible on stopping," treated with ursodeoxycholic acid. Contrast this with amiodarone's chronic hepatocellular injury, which "may persist even after stopping." What underlying mechanistic difference best explains why one is more reliably reversible than the other?

HARD

- A. Both mechanisms are identical, so there should be no difference in reversibility, contradicting the described facts
- B. Anabolic steroids impair bile transporter proteins (a functional, non-destructive process) causing cholestasis that resolves once the impairing agent is removed, whereas amiodarone causes tissue accumulation with associated phospholipidosis, mitochondrial dysfunction, and fibrosis — a more structurally destructive process that may not fully resolve simply by stopping the drug
- C. Amiodarone is renally cleared, unlike anabolic steroids, explaining the reversibility difference
- D. Ursodeoxycholic acid is what makes anabolic steroid injury special; amiodarone injury has no available treatment described
- E. Anabolic steroid hepatotoxicity is actually less reversible than amiodarone's, contradicting the premise

20. A clinical trial for a new hepatotoxic drug candidate shows dose-dependent liver injury that is fully predictable across all study subjects with a brief, consistent latent period. Based on the lecture's classification framework, and considering the described consequence that DILI is the leading cause of market withdrawal, what would be the most defensible regulatory-risk implication of this specific toxicity profile compared to an idiosyncratic profile?

HARD

- A.** An intrinsic, predictable, dose-dependent toxicity profile is regulatorily "easier to manage" in principle (via dose-limiting/monitoring strategies) compared to an unpredictable idiosyncratic profile, which cannot be reliably mitigated by dose adjustment alone and is harder to detect in limited pre-approval trial populations
- B.** Intrinsic and idiosyncratic toxicity profiles carry identical regulatory risk with no meaningful distinction
- C.** Idiosyncratic toxicity is always easier to manage than intrinsic toxicity, the reverse of the correct reasoning
- D.** Predictable, dose-dependent toxicity always leads to immediate drug withdrawal regardless of any mitigation strategy
- E.** This scenario has no bearing on regulatory risk assessment

✓ Answer key

Q	Answer	Difficulty	Why
1	A	EASY	Intrinsic hepatotoxicity is dose-dependent and predictable; idiosyncratic has no dose relationship and is unpredictable.
2	B	EASY	Acetaminophen causes zonal necrosis; N-acetylcysteine is its specific antidote.
3	B	EASY	The mainstay of DILI treatment is removing the offending agent; prognosis is usually favourable after discontinuation.
4	B	EASY	Hepatocellular injury raises ALT/AST; cholestatic injury raises ALP.
5	B	MEDIUM	Aspirin given during a viral illness in a child causing mitochondrial dysfunction/microvesicular steatosis is classic Reye's syndrome.
6	B	MEDIUM	Amoxicillin-clavulanate is the prototype idiosyncratic cholestatic hepatotoxin via clavulanic acid, producing jaundice, pruritus, and a disproportionate ALP rise.
7	A	MEDIUM	Amiodarone accumulates in the liver (phospholipidosis, mitochondrial dysfunction) causing chronic injury/fibrosis that "may persist even after stopping" — an exception to the generally favourable post-discontinuation prognosis for most DILI.
8	B	MEDIUM	This matches the hypersensitivity idiosyncratic pattern: immunologic, onset 1-5 weeks, recurs with rechallenge, rash/fever/joint pain — classic for sulfonamides.
9	B	MEDIUM	Isoniazid is metabolised to acetylhydrazine, a toxic CYP450 product causing idiosyncratic hepatocellular injury with genetic susceptibility — consistent with unpredictable transaminase elevation requiring monitoring rather than dose-based reasoning.
10	B	MEDIUM	Methotrexate increases homocysteine, causing oxidative stress that activates hepatic stellate cells, driving fibrosis.
11	B	MEDIUM	Sustained-release niacin has the highest hepatotoxic potential among antihyperlipidemics; statins rarely cause significant injury though asymptomatic transaminase rises are common.
12	B	MEDIUM	Azathioprine is linked to nodular regenerative hyperplasia (NRH), which can cause non-cirrhotic portal hypertension.
13	B	MEDIUM	The lecture lists race/ethnicity among DILI risk factors, with described differing predominant culprit drugs across populations, alongside age, sex, and genetic CYP polymorphisms as other risk factors.
14	B	MEDIUM	Tetracycline impairs mitochondrial fatty-acid oxidation causing microvesicular steatosis; valproic acid impairs mitochondrial β -oxidation plus oxidative stress causing triglyceride accumulation — both share the mitochondrial fatty-acid oxidation impairment theme.
15	B	HARD	Features like rash, fever, and eosinophilia point to an immune-mediated component (matching the hypersensitivity idiosyncratic pattern), consistent with the final "triggered immune reactions" stage of the described mechanistic cascade, as opposed to early direct cell stress or isolated mitochondrial impairment without immune involvement.
16	B	HARD	The described discriminators (dose relationship, predictability, latent period, injury pattern) provide a systematic framework for narrowing the causative agent in polypharmacy DILI, combined with temporal correlation — rather than assuming one drug is responsible without applying this framework.
17	B	HARD	Since the mechanism is progressive (glutathione depletion $\rightarrow$ NAPQI accumulation $\rightarrow$ necrosis), earlier NAC administration intervenes before glutathione is fully depleted and before extensive NAPQI-mediated injury

			has occurred, while late administration means contending with a larger toxic burden and potentially already-established cell death.
18	B	HARD	Since most DILI is asymptomatic, yet it remains the leading cause of drug withdrawals, this implies that detection substantially relies on routine laboratory monitoring (transaminases, ALP) rather than symptom-triggered reporting — an important practical implication for surveillance intensity in drug development and post-marketing follow-up.
19	B	HARD	Anabolic steroids impair bile transporter proteins — a functional disruption that can resolve once the offending drug is cleared — while amiodarone causes tissue accumulation with phospholipidosis, mitochondrial dysfunction, and fibrosis, a more structurally destructive process that may persist even after discontinuation, explaining the differing reversibility.
20	A	HARD	A predictable, dose-dependent (intrinsic) toxicity profile can in principle be managed through dosing limits and monitoring strategies, whereas idiosyncratic toxicity — being unpredictable and not dose-related — is harder to detect in limited pre-approval trial populations and cannot be reliably mitigated by dose adjustment, making it a comparatively greater regulatory and post-marketing risk, consistent with DILI's described role as the leading cause of market withdrawal.

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Asst. Prof. Nidhal Abdulkader & Asst. Lect. Nermin Jamal's DILI lecture.
Practice questions — cross-check every answer against the lecture and summary.